

25-26-2 学期概率论与数理统计 A 复习卷（一）答案

一、单项选择题

1. D 2. D 3. A 4. B 5. C 6. A

二、填空题

1. 0.2 2. $\frac{5}{100}$ 3. 0 4. $\frac{1}{4}$
 5. $\frac{3}{8}$ 6. $\frac{3}{2}$ 7. 1 8. $\frac{\bar{X}}{5}$

三、计算题

1. 解：（1） X 的分布律

X	-2	0	2
p	$\frac{1}{45}$	$\frac{16}{45}$	$\frac{28}{45}$

$$(2) E(X) = (-2) \times \frac{1}{45} + 0 \times \frac{16}{45} + 2 \times \frac{28}{45} = \frac{6}{5}$$

$$2. \text{解：(1) } P(X=1|Y=0) = \frac{P(X=1, Y=0)}{P(Y=0)} = \frac{b}{\frac{2}{25} + b} = \frac{3}{5}$$

$$\Rightarrow b = \frac{3}{25}$$

$$a + b + \frac{8}{25} = 1 \Rightarrow a = \frac{14}{25}$$

(2)

X	0	1
p	$\frac{17}{25}$	$\frac{8}{25}$

Y	0	1	2
p	$\frac{5}{25}$	$\frac{17}{25}$	$\frac{3}{25}$

$$\because P(X=0, Y=0) \neq P(X=0)P(Y=0)$$

$\therefore X, Y$ 不独立

四、解答题

1. 解：（1）平面区域 G 的面积 $S_G = 1 \times 1 = 1$

$$\therefore f(x, y) = \begin{cases} 1, & (x, y) \in G \\ 0, & \text{other} \end{cases}$$

$$(2) \quad P(X < \frac{1}{2}) = \frac{S_{\text{有效区域}}}{S_G} = \frac{\frac{1}{2} \times 1}{1} = \frac{1}{2}$$

$$(3) \quad P(Y < \frac{2}{3} | X < \frac{1}{2}) = \frac{P(Y < \frac{2}{3}, X < \frac{1}{2})}{P(X < \frac{1}{2})}$$

$$= \frac{\frac{1}{2} \times \frac{2}{3}}{\frac{1}{2}} = \frac{2}{3}$$

$$2. \text{ 解: (1) 似然函数: } L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \prod_{i=1}^n [(\theta - 2)x_i^{\theta-3}] = (\theta - 2)^n (\prod_{i=1}^n x_i)^{\theta-3}$$

$$(2) \text{ 两边取对数: } \ln L(\theta) = \ln(\theta - 2)^n + \ln(\prod_{i=1}^n x_i)^{\theta-3} = n \ln(\theta - 2) + (\theta - 3) \ln(\prod_{i=1}^n x_i)$$

$$(3) \text{ 求导: 令 } \frac{d \ln L(\theta)}{d\theta} = \frac{n}{\theta - 2} + \ln(\prod_{i=1}^n x_i) = 0$$

$$(4) \quad \theta \text{ 的极大似然估计值为 } \hat{\theta} = 2 - \frac{n}{\ln(\prod_{i=1}^n x_i)}$$

$$\theta \text{ 的极大似然估计量为 } \hat{\theta} = 2 - \frac{n}{\ln(\prod_{i=1}^n X_i)}$$

五、应用题

1. 解: 设 $A_1 = \{\text{选择甲快递}\}$, $A_2 = \{\text{选择乙快递}\}$, $A_3 = \{\text{选择丙快递}\}$,
 $B = \{\text{该店发货准时送达}\}$

$$(1) \quad (1) \quad P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) \\ = 0.5 \times 0.9 + 0.8 \times 0.3 + 0.95 \times 0.2 = 0.88$$

$$(2) \quad P(A_1|B) = \frac{P(A_1B)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P(B)}$$

$$= \frac{0.5 \times 0.9}{0.88} = \frac{45}{88}$$

(注意:全概率, Bayes 公式不写, 只有代入计算过程, 公式分值不给)

2. 解: (1) 该品牌洗衣机的平均使用寿命为 10 年

$$\begin{aligned}(2) \quad P(X > 10.1) &= 1 - P(X \leq 10.1) = 1 - P\left(\frac{X-10}{0.1} \leq \frac{10.1-10}{0.1}\right) \\ &= 1 - \Phi(1) \\ &= 1 - 0.8413 = 0.1587 \\ (3) \quad P(9.8 < X < 10.2) &= P\left(\frac{9.8-10}{0.1} < \frac{X-10}{0.1} < \frac{10.2-10}{0.1}\right) = \Phi(2) - \Phi(-2) \\ &= 2\Phi(2) - 1 = 2 \times 0.9772 - 1 = 0.9544\end{aligned}$$

六、证明题

$$(1) \text{ 解: } P(X > 1) = 1 - P(X \leq 1) = 1 - F(1) = e^{-2}$$

$$\therefore F(1) = 1 - e^{-2}$$

$$\therefore X \sim E(\lambda)$$

$$\therefore F(x) = \begin{cases} 1 - e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad \text{且 } \lambda = 2$$

$$(2) \quad \because E(X) = \frac{1}{2}, \quad D(X) = \frac{1}{2^2}$$

$$\therefore \frac{E(X)}{D(X)} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

25-26-2 学期概率论与数理统计 A 复习卷 (二) 答案

一、单项选择题

1. D 2. A 3. B 4. B 5. C 6. D

二、填空题

1. $\frac{3}{10}$ 2. 36 3. $\frac{1}{9}$ 4. 0.85
5. 1 6. $\frac{1}{6}$ 7. $\frac{1}{4}$ 8. 1

三、计算题

解：(1) Y 的分布律

Y	1	2	4
p	0.5	0.25	0.25

$$(2) E(Y^2) = 1^2 \times 0.5 + 2^2 \times 0.25 + 4^2 \times 0.25 = 5.5$$

2. 解：(1) $P(X=0) = a + 0.1 + 0.2 = 0.5 \Rightarrow a = 0.2$

$$a + 0.1 + 0.2 + 0.1 + b + 0.2 = 1 \Rightarrow b = 0.2$$

(2)

X	0	1
p	0.5	0.5

Y	-1	0	1
p	0.3	0.3	0.4

$$E(X) = 0 \times 0.5 + 1 \times 0.5 = 0.5$$

$$E(Y) = -1 \times 0.3 + 0 \times 0.3 + 1 \times 0.4 = 0.1$$

$$E(XY) = 1 \times (-1) \times 0.1 + 1 \times 1 \times 0.2 = 0.1 \quad \text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= 0.1 - 0.1 \times 0.5 = 0.05$$

故 X, Y 相关

四、解答题(共 16 分, 每小题 8 分)

1. 解：(1) 平面区域 D 的面积 $S_D = \pi$

$$\therefore f(x, y) = \begin{cases} \frac{1}{\pi}, & (x, y) \in D \\ 0, & \text{other} \end{cases}$$

$$(2) P(Y > 0) = \frac{\frac{1}{2} \times \pi}{\pi} = \frac{1}{2}$$

$$(3) P(X > 0 | Y > 0) = \frac{P(X > 0, Y > 0)}{P(Y > 0)}$$

$$= \frac{\frac{1}{4} \pi}{\frac{1}{2} \pi} = \frac{1}{2}$$

3. 解: (1) 似然函数: $L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \prod_{i=1}^n \frac{\theta}{x_i^{\theta+1}} = \frac{\theta^n}{(\prod_{i=1}^n x_i)^{\theta+1}}$

(2) 两边取对数: $\ln L(\theta) = \ln \theta^n - \ln(\prod_{i=1}^n x_i)^{\theta+1} = n \ln \theta - (\theta+1) \ln(\prod_{i=1}^n x_i)$

(3) 求导: 令 $\frac{d \ln L(\theta)}{d\theta} = \frac{n}{\theta} - \ln(\prod_{i=1}^n x_i) = 0$

(4) θ 的极大似然估计值为 $\hat{\theta} = \frac{n}{\ln(\prod_{i=1}^n x_i)}$

θ 的极大似然估计量为 $\hat{\theta} = \frac{n}{\ln(\prod_{i=1}^n X_i)}$

六、应用题

2. 解: 设 $A_1 = \{\text{偏胖者}\}$, $A_2 = \{\text{不胖不瘦者}\}$, $A_3 = \{\text{偏瘦者}\}$,

$B = \{\text{成年居民患高血压}\}$

(1) $P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$

$= 0.1 \times 0.2 + 0.8 \times 0.1 + 0.1 \times 0.05 = 0.105$

(2) $P(A_1|B) = \frac{P(A_1 B)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P(B)}$

$= \frac{0.1 \times 0.2}{0.105} = \frac{4}{21}$

(注意: 全概率, Bayes 公式不写, 只有代入计算过程, 公式分值不给)

2. 解: (1) 该高校男生的平均身高为 1.7m

(2) $P\{X > 1.86\} = 1 - P\{X \leq 1.86\}$

$= 1 - \Phi\left(\frac{1.86 - 1.7}{0.08}\right)$

$= 1 - \Phi(2)$

$= 1 - 0.9772 = 0.0228$

(3) $P(1.62 \leq X \leq 1.78) = \Phi\left(\frac{1.78 - 1.7}{0.08}\right) - \Phi\left(\frac{1.62 - 1.7}{0.08}\right)$

$= \Phi(1) - \Phi(-1)$

$$= 2\Phi(1) - 1$$

$$= 2 \times 0.8413 - 1 = 0.6826$$

七、证明题

证明: (1) $\because X \sim P(1)$, 由 $P(X=1) = e^{-\lambda} \frac{\lambda^1}{1!}$, $P(X=2) = e^{-\lambda} \frac{\lambda^2}{2!}$

$$\therefore e^{-\lambda} \frac{\lambda^1}{1!} = e^{-\lambda} \frac{\lambda^2}{2!}$$

$$\therefore \lambda = 2$$

$$(2) E(X) = 2, D(X) = 2$$

$$\because D(X+2) = D(X) = 2$$

$$\therefore \frac{D(X+2)}{E(X)} = \frac{2}{2} = 1$$

25-26-2 学期概率论与数理统计 A 复习卷 (三) 答案

一、单项选择题

1. B 2. B 3. C 4. D 5. A 6. C

二、填空题

1. $\frac{1}{14}$ 2. 11 3. $\frac{1}{3}$ 4. 1
5. $\frac{1}{2}$ 6. $\frac{1}{4}$ 7. 0.3 8. 6

三、计算题

1. 解: (1) $c=0.1$

(2) Y 的分布律

Y	1	2	5
p	0.4	0.3	0.3

$$(3) P(Y < 3) = P(Y=1) + P(Y=2) = 0.4 + 0.3 = 0.7$$

2. 解: (1) $E(X) = -1 \times \frac{1}{4} + 0 \times \frac{1}{2} + 1 \times \frac{1}{4} = 0$

$$E(Y) = 0 \times \frac{1}{2} + 1 \times \frac{1}{2} = \frac{1}{2}$$

$$E(XY) = 0$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0$$

(2) X, Y 的联合分布律为

$X \backslash Y$	0	1
-1	$\frac{1}{4}$	0
0	0	$\frac{1}{2}$
1	$\frac{1}{4}$	0

四、解答题

1. 解: (1) 平面区域 G 的面积 $S_G=4$

$$\therefore f(x, y) = \begin{cases} \frac{1}{4}, & (x, y) \in G \\ 0, & \text{other} \end{cases}$$

$$(2) P(X < 1) = \frac{1 \times 2}{4} = \frac{1}{2}$$

$$(3) P(Y < 1 | X < 1) = \frac{P(X < 1, Y < 1)}{P(X < 1)}$$

$$= \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$

解: 2. 解: (1) 似然函数: $L(\theta) = \prod_{i=1}^n f(x_i, \theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_{i=1}^n x_i}$

$$(2) \text{两边取对数: } \ln L(\theta) = n \ln \theta - \theta \sum_{i=1}^n x_i$$

$$(3) \text{求导: 令 } \frac{d \ln L(\theta)}{d\theta} = \frac{n}{\theta} - \sum_{i=1}^n x_i = 0$$

$$(4) \theta \text{ 的极大似然估计值为 } \hat{\theta} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

θ 的极大似然估计量为 $\hat{\theta} = \frac{n}{\sum_{i=1}^n X_i} = \frac{1}{\bar{X}}$

五、应用题

1. 解：设 A_1 表示小华从福州到广州乘坐汽车， A_2 表示小华从福州到广州乘坐高铁， A_3 小华从福州到广州乘坐飞机， A_1, A_2, A_3 构成完备事件组，

设 B 表示小华从福州到广州迟到，

则 $P(A_1) = 0.2, P(A_2) = 0.3, P(A_3) = 0.5, P(B|A_1) = 0.2, P(B|A_2) = 0.1, P(B|A_3) = 0.04$

$$(1) \quad P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$= 0.2 \times 0.2 + 0.3 \times 0.1 + 0.5 \times 0.04 = 0.09$$

$$(2) \quad P(A_1|B) = \frac{P(A_1)P(B|A_1)}{P(B)} = \frac{P(A_1)P(B|A_1)}{P(B)}$$

$$= \frac{0.2 \times 0.2}{0.09} = \frac{4}{9}$$

(注意：全概率, Bayes 公式不写，只有代入计算过程，公式分值不给)

2. 解：(1) 人的智力商数 (IQ) 的平均值为 100

$$(2) \quad P\{X > 130\} = 1 - P\{X \leq 130\}$$

$$= 1 - \Phi\left(\frac{130 - 100}{15}\right)$$

$$= 1 - \Phi(2)$$

$$= 1 - 0.9772 = 0.0228$$

$$(3) \quad P(85 \leq X \leq 115) = \Phi\left(\frac{115 - 100}{15}\right) - \Phi\left(\frac{85 - 100}{15}\right)$$

$$= \Phi(1) - \Phi(-1)$$

$$= 2\Phi(1) - 1$$

$$= 2 \times 0.8413 - 1 = 0.6826$$

六、证明题

证明：解：(1) $f(x) = \begin{cases} \frac{1}{6}, & 0 < x < 6 \\ 0, & \text{other} \end{cases}$

$$(2) \quad \text{由 } X \sim U(0, 6) \quad \therefore E(X) = 3, \quad D(X) = 3$$

$$\therefore E(X^2) = D(X) + [E(X)]^2 = 3 + 3^2 = 12$$

$$E[X(X-5)] = E(X^2 - 5X) = E(X^2) - 5E(X) = 12 - 5 \times 3 = -3$$